



Synthesis and Nuclear Radiation Shielding Ability of $\text{Li}_2\text{O}-\text{ZnO}-\text{P}_2\text{O}_5$ Glasses: The Role of Yb_2O_3

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Abstract

Nuclear radiation shielding parameters for glass samples composed of 20 mol.% Li_2O -40 mol.% ZnO -40 mol.% P_2O_5 : x wt.% Yb_2O_3 ($x=0, 0.2, 0.4, 0.6, 0.8, 1, 2$) were evaluated. The samples were synthesized using the melt-quenching technique. The prepared samples were encoded Yb0.0, Yb0.2, Yb0.4, Yb0.6, Yb0.8, Yb1 and Yb2 according to the Yb_2O_3 content. X-ray diffraction (XRD) reveals that all the prepared samples were amorphous. The gamma ray shielding parameters such as mass attenuation coefficient (MAC), linear attenuation coefficient (LAC), half value layer (HVL) and mean free path (MFP) were obtained using Phy-X/PSD and XCOM software programs. The atomic cross section (ACS), electronic cross section (ECS), effective atomic number (Z_{eff}), and electron density (N_{eff}) were evaluated, as well as exposure buildup factor (EBF) and energy absorption buildup factor (EABF). The MAC values were found to be 26.304–31.872 cm^2/g for Yb0.0–Yb2, respectively. The HVL and MFP values were found to be (0.008–0.007 cm) and (0.012–0.010 cm) for Yb0.0–Yb2, respectively. The ACS and ECS values were found to be (9.912×10^{-21} cm^2/g) up to 13.06×10^{-21} cm^2/g and (3.990×10^{-23} up to 4.900×10^{-23} cm^2/g) for Yb0.0–Yb2, respectively. The Z_{eff} and N_{eff} values were found to be (24.83–26.64) and (6.59×10^{23} up to 6.50×10^{23} electrons/g) for Yb0.0–Yb2, respectively. The results showed that the addition of Yb_2O_3 improves gamma-ray shielding properties of the glass. On the other hand, adding Yb_2O_3 to the prepared glass leads to a decrease in the removal cross section for fast neutrons (Σ_R) from 0.103 to 0.096 cm for Yb0.0–Yb2, respectively. The gamma-ray shielding parameters for the present samples were compared with commercial concrete and glass used as gamma ray shields.

Keywords Phosphate glass · gamma ray shields · Yb · nuclear parameters

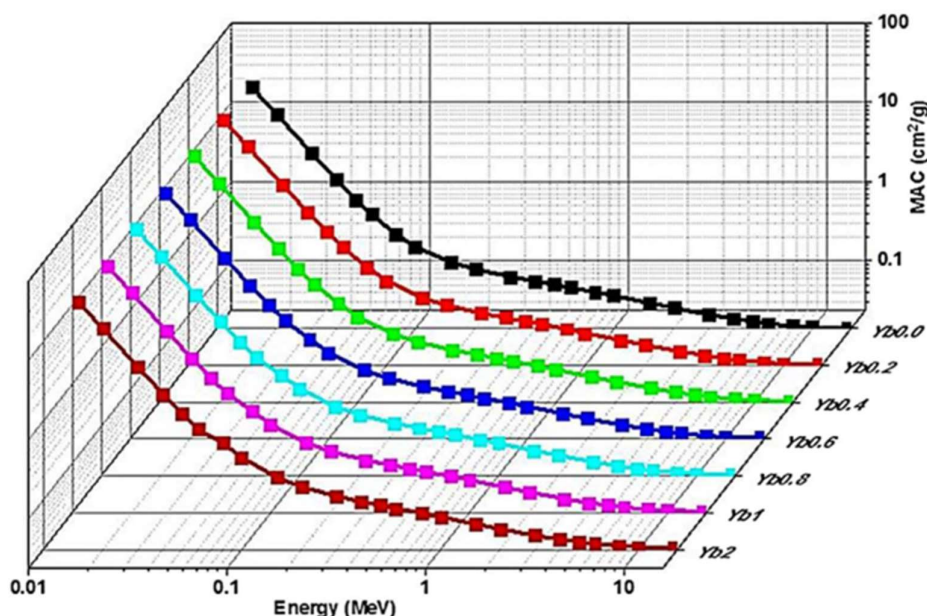


Fig. 3 MAC versus photon energy for all glass samples.

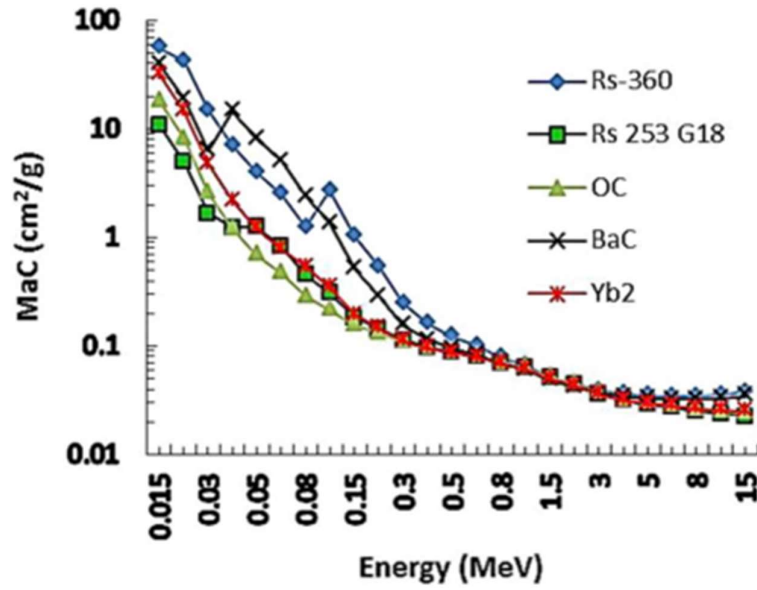


Fig.4 Comparison of the minimum and maximum values of the MAC of the Yb2 and commercial materials.

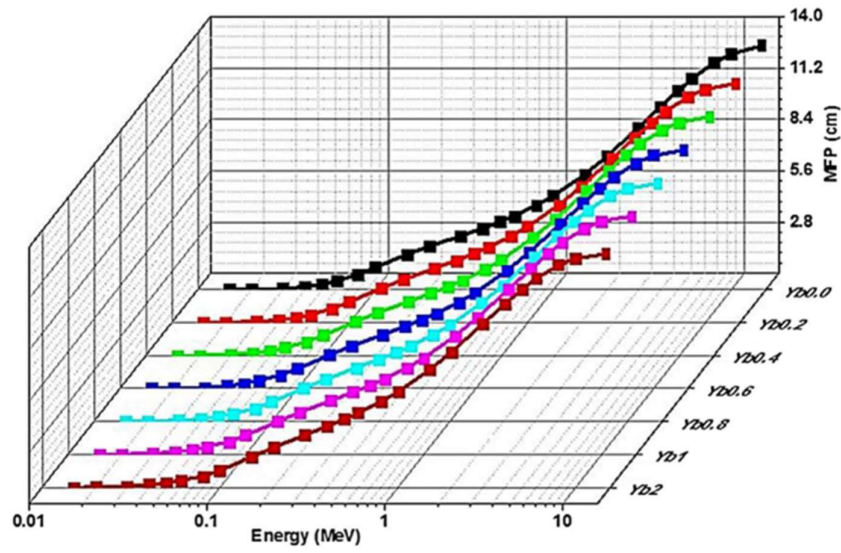


Fig. 8 MFP versus photon energy for all glass samples.

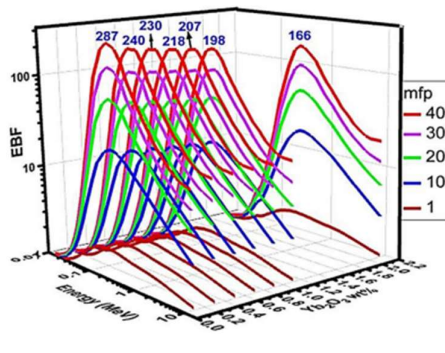


Fig. 14 Exposure buildup factor (EBF) versus photon energy for all glass samples.

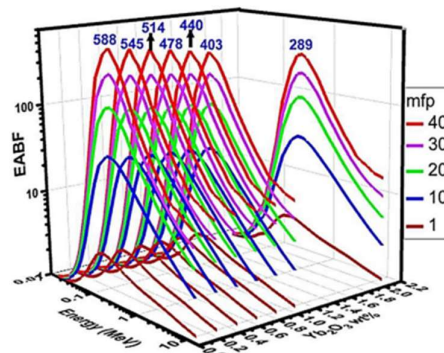


Fig.15 Energy absorption buildup factor (EABF) versus photon energy for all glass samples.